

FedBridge: Accelerating Edge-Assisted Federated Learning for Model-Heterogeneous Clients

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Abstract

In an edge-assisted federated learning (FL) system, edge servers aggregate client models into intermediate models for the cloud server to produce the global model communication-efficiently. However, existing edge-assisted systems fail to accommodate model-heterogeneous clients, i.e., clients running models with different architectures. To tackle this problem, this paper proposes FedBridge, a novel edge-assisted system that enables FL across model-heterogeneous clients through a two-tier knowledge-sharing mechanism. It deploys an expandable fusion model on each edge server in the system to fuse the knowledge from heterogeneous client models through knowledge distillation. In the meantime, it employs a contrastive loss to mitigate data heterogeneity in client data by aligning the logits of the fusion model close to those of the global model. On the cloud server, it employs a block-based aggregation method to merge fusion models transmitted from the edge servers. We conduct extensive experiments with three models on two widely-used public datasets to evaluate the performance of FedBridge. The results demonstrate that, compared to state-of-the-art systems, FedBridge accelerates model convergence by up to 6.3x and improves model accuracy by 6.2%-17.7%, with 89.51%-96.13% reduction in communication overhead and 48.12%-61.77% in memory overhead.

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CCS Concepts

• **Computing methodologies** → **Machine learning**; **Parallel computing methodologies**; • **Human-centered computing** → **Ubiquitous and mobile computing**.

Keywords

Edge-assisted FL, logit-contrastive loss, heterogeneous model, data heterogeneity

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1 Introduction

Edge devices, including smartphones and Web-of-Things (WoT) devices, account for over half of global Internet traffic and generate massive, heterogeneous data that power machine learning (ML) applications such as recommender systems [54] and social networks [55]. The ML models powering these applications rely on private data collected or produced by mobile and WoT devices, leading to significant privacy concerns. Federated Learning (FL) [17, 22, 31, 37] has gained significant attention as a promising approach for training ML models in a privacy-preserving manner across distributed edge devices (often referred to as *clients*). FL enables clients to collaboratively train a shared global ML model with the coordination of a cloud server [8, 13, 37] or AllReduce [25, 30, 51]. In each training round, clients independently train their models on their private data and transmit them to the cloud server. The server aggregates these client models into a global model by aggregating their parameters with methods like FedAvg [5, 27, 37] and then distributes the global model to clients for the next training round.

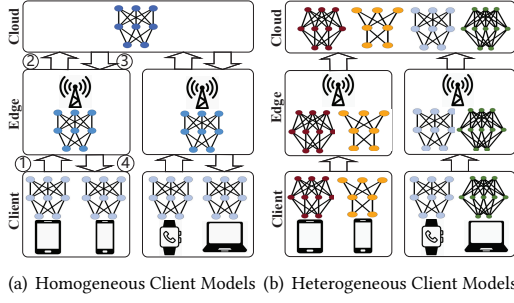


Figure 1: Edge-assisted FL system: homogeneous client models vs. heterogeneous client models.

This iterative process effectively fuses clients' knowledge into the global model while preserving their data privacy.

The transmission of models between clients and the cloud server incurs significant communication overhead [9, 32, 45, 46]. For example, training a CNN model with three convolutional layers on MNIST dataset across 50 clients produces a total of 8,250MB of network traffic [45]. This communication overhead is becoming prohibitively expensive as the sizes of modern ML models increase rapidly in pursuit of high performance [10, 35, 45]. In recent years, researchers have begun to leverage the unique advantages of edge computing to alleviate the communication overhead in FL systems [11, 36, 45, 50]. Edge computing offers near-user computational and storage resources by deploying edge servers within close geographic proximity to users [1, 38, 40]. An edge-assisted FL system can be built across a cloud server and multiple edge servers [45]. As illustrated in Fig. 1, in an edge-assisted FL system, clients transmit their models to nearby edge servers, e.g., those attached to base stations or deployed in regional data centers. These edge servers aggregate client models into intermediate models and transmit them to the cloud server for aggregation. By transmitting only intermediate models to the cloud server, edge-assisted FL can reduce the communication overhead by up to 94.3% [11, 45, 48].

Most existing edge-assisted FL systems assume that all clients in the system run models with identical architectures. In the real world, clients are model-heterogeneous - they run models tailored to their hardware capacities and/or software needs [44, 52]. For example, a flagship mobile phone with luxury memory resources may run a ViT [7, 42] to power its CV functionality, while an old one can only afford to run a ResNet-8. Mobile apps also often leverage heterogeneous models to perform the same AI tasks. A recent study of 267 popular Android apps revealed that 83.78% of them are powered by heterogeneous models offering the same AI functionality [19].

Existing edge-assisted FL systems fail to accommodate these model-heterogeneous clients because edge servers cannot aggregate their models with methods like FedAvg [37, 45]. A popular solution to model heterogeneity in FL systems is knowledge distillation (KD) [3, 28, 34]. In KD-based FL systems, clients send their models to the cloud server in every training round so that the cloud server can train a global model by distilling knowledge from these client models. However, this approach reintroduces the issue of communication inefficiency.

This paper presents FedBridge, a novel edge-assisted FL system designed to enable communication-efficient knowledge sharing across model-heterogeneous clients. In FedBridge, the clients first train their models and send them to nearby edge servers. Unlike conventional FL systems, where edge servers aggregate client models into intermediate models, FedBridge deploys a *fusion* model on each edge server in the system to distill and integrate the knowledge from clients' potentially heterogeneous client models (§4.2). After that, edge servers send these fusion models to the cloud server for aggregation (§4.3). Next, the cloud server aggregates received fusion models to generate the global model for transmission to the edge servers for updating their fusion models and client models. The main contributions of this paper are fivefold:

- To our best knowledge, FedBridge is the first edge-assisted FL systems that enables communication-efficient knowledge sharing across model-heterogeneous clients. Compared with existing FL systems, including edge-assisted ones, it allows a much larger scale and higher variety of clients to share their knowledge practically in a federated manner.
- FedBridge deploys fusion models on edge servers to fuse knowledge from client models. They can scale up on demand to accommodate as they learn more from client models. This improves model accuracy, accelerates model convergence, and reduces the memory overheads on edge servers.
- FedBridge introduces a logit-contrastive loss during the training of fusion models on edge servers. This effectively accommodates the data heterogeneity across clients, which is more likely to occur with an increased scale and variety of clients in the system.
- FedBridge employs a new method for the cloud server to aggregate fusion models of different sizes by averaging their common modules. This new model aggregation allows the cloud server to effectively integrate the knowledge from fusion models into the global model.
- Extensive experiments are conducted with three ML models on two widely-used public datasets to evaluate FedBridge. The results show that, compared with state-of-the-art edge-assisted FL systems, FedBridge improves model accuracy by 6.2%-17.7%, and accelerates model convergence by up to 6.3x.

2 Background

2.1 Edge-Assisted FL

An edge-assisted FL system consists of a cloud server C , a set of M edge servers $S = \{s_1, \dots, s_M\}$, and a set of clients $U = \{u_1, \dots\}$ located in the coverage areas of these edge servers [36, 45, 49]. Each edge server $s_m \in S$ covers a subset of U , and each client $u \in U$ trains a client model w_L on its dataset $\mathcal{D}_u = \{(x_i, y_i)\}_{i=1}^{|\mathcal{D}_u|}$. Fig. 1(a) shows an example edge-assisted FL system that involves clients, edge servers, and the cloud server. Each FL training round goes through four steps. ① Client Training: Clients train their local models on their data and send them to the corresponding edge servers. ② Edge Aggregation: Edge servers aggregate client models to produce intermediate models and transmit these intermediate models to the cloud for the production of the global model. ③ Cloud Model Aggregation: The cloud server aggregates intermediate models into a global model for distribution to the edge servers. ④ Model Updates. Based on the received global model, edge servers update

their intermediate models and client models. Next, they send updated client models to corresponding clients for the next training round. This process iterates until the stop criterion is met, e.g., the global model reaches a target accuracy. Fig. 1(a) illustrates the transmission process in detail.

2.2 Heterogeneity in FL Systems

Knowledge distillation (KD) enables the transfer of knowledge between architectural-heterogeneous DNN models [12, 18, 39]. Typically, knowledge transfer involves utilizing a pre-trained large teacher model w_T to guide the learning of a small student model w_S [12, 39]. The fundamental principle is to optimize w_S so that its output logits closely resemble those generated by w_T . Generally, the Kullback-Leibler (KL) divergence loss is employed to train w_S :

$$\min_{w_S} \mathbb{E}_{x_i \sim \mathcal{D}} (D_{KL}(\sigma(f(w_T; x_i)) \parallel \sigma(f(w_S; x_i)))) \quad (1)$$

where $f(w_T; x_i)$ and $f(w_S; x_i)$ represent the output logits of w_T and w_S on sample x_i , respectively, σ denotes the softmax function, and D_{KL} represents the KL divergence. Researchers have utilized KD to enable FL across model-heterogeneous clients [3, 28, 34]. In a KD-based FL system, clients transmit their models to the cloud server. The cloud server then aggregates the logits generated by these models on a pre-prepared unlabeled dataset and trains the global model with the aggregated logits.

2.3 Contrastive Learning.

Contrastive learning aims to create a representation space where similar instances are close together and dissimilar ones are far apart [2, 4]. The key idea is to reduce the distance between positive data pairs and increase the distance between negative data pairs within the embedding space [2]:

$$l_{i,j} = -\log \frac{\exp(\text{sim}(x_i, x_j)/\tau)}{\sum_{k=1}^{2N} \exp(\text{sim}(x_k, x_i)/\tau)} \quad (2)$$

where x_i and x_j represent two distinct views of the same image, N is the number of images, $\text{sim}(\cdot)$ denotes the cosine similarity function, and τ is a temperature parameter. The loss is computed by aggregating the contrastive losses across all image pairs. Unlike traditional contrastive learning that compares the representations of images, FedBridge performs contrastive training by comparing the logits produced models, including the global model, fusion models, and client models (§4.2).

3 Motivation

Communication Overheads. Existing edge-assisted FL systems have commonly assumed that all client models are architecturally identical. Only under this assumption can edge servers aggregate these client models into intermediate models by averaging their model parameters, as shown in Fig. 1(a). However, in the real world, clients' edge devices often need to run DNN models that suit their hardware capacities and software limitations. For example, a smartphone with 8GB memory can only afford to run a ResNet-18 while one with 16GB memory can accommodate a ResNet-152. This model heterogeneity prevents edge servers from fusing client models into intermediate models by directly averaging their parameters. Instead, edge servers must transmit heterogeneous client models to

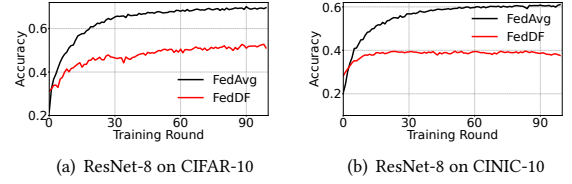


Figure 2: Comparison in model convergence between FedAvg [37] and FedDF [34]. In both FedAvg and FedDF, every client trains a ResNet-8 model [16] on the same dataset to ensure comparability and fairness - FedAvg cannot accommodate heterogeneous client models. Note: When integrating knowledge from heterogeneous client models, FedDF performs even worse than in this experiment.

the cloud server for knowledge aggregation, as shown in Fig. 1(b). Handling heterogeneous client models in this way, the edge-assisted FL system will incur tremendous network traffic and fall short of its promised communication efficiency.

Model Heterogeneity. Although knowledge distillation makes it possible to transfer knowledge across heterogeneous models, KD-based FL systems struggle to effectively integrate the knowledge from client models into the global model. The fundamental reason is that logits cannot fully capture and represent the knowledge embedded in client models. To validate this, we conduct an experiment to compare the performance of FedDF [34], a classic KD-based FL system, and FedAvg, a conventional FL system based on model aggregation [37]. In FedAvg, the cloud server averages clients' homogeneous models to produce the global model. In FedDF, the cloud server trains a global model of the same architecture as client models by aggregating the logits produced by client models. Fig. 2 compares the convergence of the global model in FedAvg and FedDF on CIFAR-10 and CINIC-10. We can see the significant performance gap between FedDF and FedAvg. For example, as shown in Fig. 2(a), FedAvg converges the model to an accuracy of 52.79% on CIFAR-10 within just 12 training rounds, while FedDF requires 98 training rounds to achieve the same accuracy. In addition, FedAvg achieves a top-1 accuracy of 70.12% on CIFAR-10, significantly outperforming FedDF, which only reaches 52.79%. The results reveal that knowledge distillation pays a high performance price for its ability to accommodate model heterogeneity in FL systems. Thus, it cannot be applied off the shelf to edge-assisted FL systems to accommodate the model heterogeneity in client models. To tackle this challenge, FedBridge deploys fusion models on edge servers to facilitate effective knowledge fusion from client models (§4.2) and support knowledge integration into the global model (§4.3).

Data Heterogeneity. There is also data heterogeneity across the clients in an FL system. The data used by clients to train their models often differ in distribution [10, 14, 20, 24, 56]. This poses a grand challenge to model aggregation. It is well known model aggregation methods such as FedAvg, which assume model homogeneity across clients, often suffer from suboptimal model performance when their data are non-independent and identically distributed (non-IID). Not surprisingly, we found that KD-based FL systems, despite their flexibility in accommodating model heterogeneity in FL systems, also suffer poor model performance in non-IID settings. Fig. 3 shows the top-1 accuracy achieved by FedDF for ResNet-8 and ResNet-56

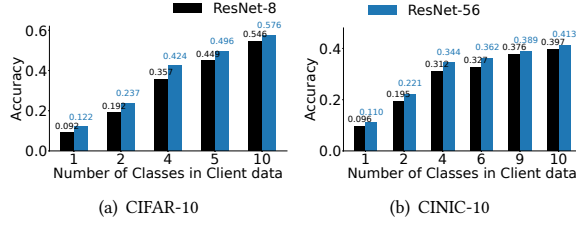


Figure 3: Top-1 model accuracy comparison in different non-IID settings, where the x axis represents the number of classes in each individual client’s data.

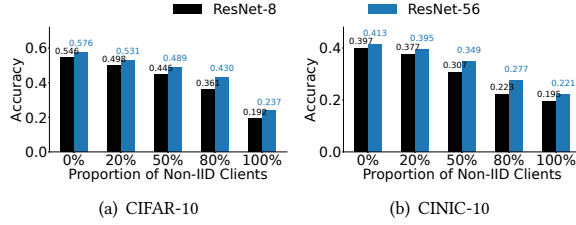


Figure 4: Top-1 model accuracy comparison in different non-IID settings, where the number of the x axis represents the proportion of non-IID clients in the FL system.

on the CIFAR-10 and CINIC-10 datasets in non-IID settings. We can see that the increase in data heterogeneity makes it remarkably harder for FedDF to achieve a high model accuracy. As introduced above, the ability of FedBridge to accommodate model heterogeneity allows a larger scale and higher variety of clients to share their knowledge communication-efficiently. This inevitably increases the possibility and the severity of data heterogeneity in the system. To evaluate its impact on system performance, we conduct an experiment where the proportion of non-IID clients increases. Fig. 4 demonstrates the results. As expected, an increased proportion of non-IID clients makes it harder for clients to converge their models to a high accuracy. To mitigate the impact of data heterogeneity, FedBridge drives fusion models close to the global model and away from client models to retain the common knowledge across clients and reduce the biases introduced by their heterogeneous data from clients.

Fixed Model Sizes. FedDF [34] also fuses the knowledge from heterogeneous client models through knowledge distillation. It trains a global model of the same size as client models. This design, by nature, limits the capacity of the global model to accommodate clients’ knowledge - a larger model can and is often needed to capture and store more knowledge extracted from training data [16, 41]. To mitigate this limitation, Fed-ET [3] trains a larger model on the cloud server. It allows Fed-ET to outperform FedDF in both model accuracy and model convergence. However, FedDF and Fed-ET both rely on training a fixed-size model on the cloud server. Our experiment reveals that the optimal model size for effective knowledge distillation varies with the number of participating clients in the federated learning process, suggesting that a fixed-size model may not always be ideal. Table 1 presents the top-1 accuracy achieved by Fed-ET for the global model of different architectures (and sizes)

Table 1: Top-1 accuracy of Fed-ET with different global models and varying numbers of clients training ResNet-8.

Model	CIFAR-10			CINIC-10		
	# of Clients			# of Clients		
	2	10	50	2	10	50
ResNet-8	0.455	0.541	0.566	0.398	0.405	0.413
ResNet-32	0.476	0.557	0.586	0.392	0.417	0.429
ResNet-56	0.469	0.576	0.591	0.388	0.409	0.435
ResNet-110	0.459	0.551	0.603	0.378	0.397	0.422

and varying numbers of clients. The results reveal two key observations: (1) Global models of different architectures and sizes yield different top-1 accuracies for the global model with the same number of clients. For example, with two clients in the system, the top-1 accuracies of the global model achieved on CIFAR-10 are 0.455, 0.476, 0.469, and 0.459 for ResNet-8, ResNet-32, ResNet-56, and ResNet-110, respectively. (2) As the number of participating clients increases, a larger global model can converge to a higher top-1 accuracy. Specifically, ResNet-32 achieves the highest accuracy with two clients in the system, whereas ResNet-56 and ResNet-110 outperform other models when the number of clients increases to 10 and 50 on CIFAR-10, respectively. Similar trends can be observed on CINIC-10. The underlying reason is that with a small number of clients (and consequently less diverse knowledge overall), a small model can better capture and fuse their knowledge, while an overly large one is prone to overfitting. Conversely, with a large number of clients in the system, a small model often struggles with underfitting, whereas a large model can generalize better. These findings highlight the significant impact of model size on FL and underscore the necessity of selecting an appropriate global model size for optimal system performance.

4 FedBridge

To reduce communication overhead, mitigate data heterogeneity, and improve model accuracy, FedBridge leverages ensemble knowledge distillation and contrastive learning to train an expandable fusion model on each edge server (§4.2), and employs a block-based aggregation strategy to merge fusion models in the cloud server (§4.3).

4.1 Overview

FedBridge is a new edge-assisted FL system that accommodates heterogeneity in client models to improve learning performance. Fig. 5 provides an overview of FedBridge. Its learning process goes through 7 steps. ① Client Model Training and Transmission: Clients train their models of potentially heterogeneous architectures on their private datasets and transmit them to corresponding edge servers. ② Fusion Model Training: Each edge server trains an expandable fusion model larger than client models based on the ensemble knowledge from client models. ③ Fusion Model Transmission: Edge servers transmit their fusion models to the cloud server. ④ Cloud Model Aggregation: The cloud server leverages a novel block-based aggregation strategy to merge received fusion models into the global model to fuse their knowledge. ⑤ Global Model Distribution: The cloud server generates the global model by

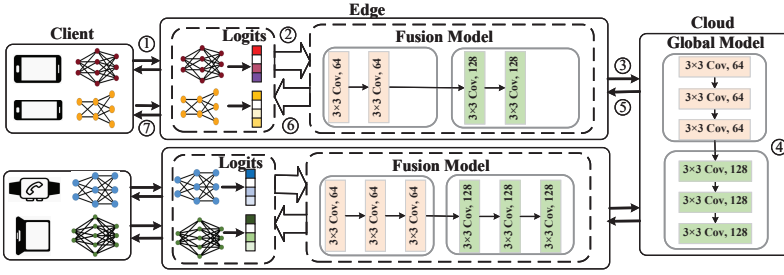


Figure 5: FedBridge overview.

aggregating received fusion models. Then, it transmits the modules of the global model to the corresponding edge servers. ⑥ Client Model Updating: Edge servers update their fusion models accordingly and distill the knowledge in these fusion models into client models. ⑦ Client Model Distribution: Edge servers send updated client models to corresponding clients. When the clients receive these models, they start the next training round.

4.2 Fusion Model Training

Ensemble Knowledge Distillation. Existing edge-assisted FL systems [45] typically assume that clients share homogeneous models, allowing edge servers to aggregate client models by simply averaging their parameters with methods like FedAvg [37]. However, in real-world scenarios, model heterogeneity prevents edge servers from directly merging client models into a single fusion model. Recent FL studies [3, 34] have explored knowledge distillation to train a global model on the cloud server. FedDF [34] constructs a global model of the same architecture as client models, while Fed-ET [3] trains a larger global model. However, Fed-ET requires all client models to have identical representation layers, limiting its applicability in heterogeneous settings. To fully accommodate model heterogeneity, FedBridge deploys and trains a fusion model at each edge server larger than client models to integrate the knowledge from client models. In this way, FedBridge significantly reduces the communication overhead incurred by model transmissions during the FL process, as edge servers only transmit their fusion models to the cloud server instead of client models.

Fig. 6 illustrates the process of fusion model training in FedBridge. During each FL training round, clients transmit their models, w_L , to their respective edge servers. Each edge server distills the ensemble knowledge from their $|s|$ client models on unlabeled data, w_L^u , into a single fusion model, w_I . For each unlabeled input x , the edge server extracts the logits from the client models, $z_L^u = f(w_L^u; x)$, and the logits from the fusion model, $z_I = f(w_I; x)$. The corresponding loss function is formulated as:

$$L_{KD} = KL \left(\sigma \left(\frac{1}{|S|} \sum_{u \in S} z_L^u \right), \sigma(z_I) \right) \quad (3)$$

where $\frac{1}{|S|} \sum_{u \in S} z_L^u$ calculates the averaged logits of client models for input x , KL denotes Kullback–Leibler divergence, and σ is the softmax function.

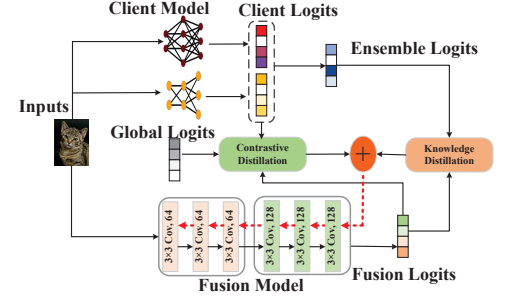


Figure 6: Fusion model training.

Contrastive Distillation. In FL systems, data heterogeneity is a prevalent challenge. We observe that ensemble knowledge distillation-based FL struggles to effectively mitigate the issues arising from heterogeneous data, as discussed in Section 2.2. In FedBridge, the heterogeneity of clients' training data directly translates into the heterogeneity of their logits, which undermines the performance of the fusion models distilled from these logits. In recent years, contrastive learning [2] has emerged as a technique for training ML models. Its core principle is to improve training performance by minimizing the outcome distance between positive data samples while maximizing the separation from negative data samples. Moon [29] made an early attempt to integrate contrastive learning into FL systems to improve system performance on non-IID data. However, Moon requires that all client models share the same architecture. Thus, it cannot accommodate heterogeneous client models. To mitigate the bias introduced by the data heterogeneity across clients, the global model in an FL system needs to capture general knowledge across client models. To achieve this goal, FedBridge leverages contrastive learning to align the fusion models to the global model. Specifically, it minimizes the discrepancy between the logits of fusion models on the unlabeled data, z_I , and those of the global model, z_G , while maximizing the divergence between the logits of fusion models, z_I , and client models, z_L^u . This is implemented with a contrastive distillation loss L_{CD} when training fusion models:

$$- \sum_{u \in S} \frac{1}{|s|} \log \left(\frac{\exp(\text{sim}(z_I^I, z_G)/\tau)}{\exp(\text{sim}(z_I, z_G)/\tau) + \exp(\text{sim}(z_I, z_L^u)/\tau)} \right) \quad (4)$$

where τ is a temperature parameter, and $\text{sim}(\cdot)$ is the cosine similarity function. Edge servers generate global logits by passing input x through w_G , i.e., the global model from the last FL training round. Fig. 7 illustrates the process of calculating the contrastive loss L_{CD} . After computing L_{CD} and L_{KD} , edge servers train their fusion models with $L = L_{KD} + \mu L_{CD}$ as the loss function, where μ is a hyperparameter that balances the contribution of the contrastive distillation loss to loss calculation.

Expandable Model for Knowledge Fusion. As discussed in Section 2.2, FedDF and Fed-ET train a fixed-size global model on the cloud server, making model performance highly dependent on its size. However, identifying an optimal model size for efficient model fusion remains a challenge.

The current prevalent DNN architectures, such as VGG [41], ResNet [16], and ViT [7], exhibit a consistent structural configuration. Each network is constructed as a cascade of its sub-networks.

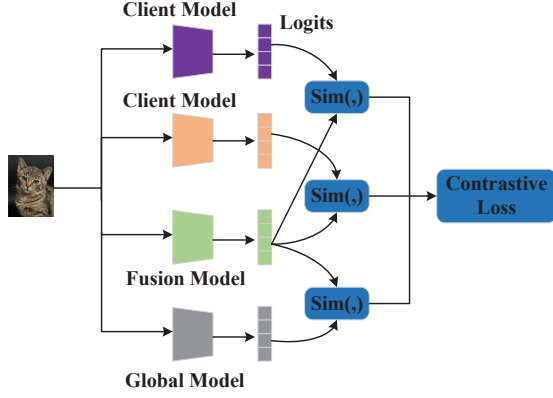


Figure 7: Contrastive distillation for training fusion models.

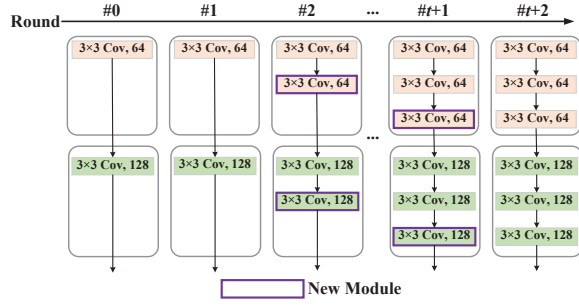


Figure 8: Fusion model expansion. This fusion model grows two new blocks in round #2 and another two in round #t + 1. Its growth stops in round #t+2 when stop criterion is met.

These sub-networks, in turn, consist of modules, which typically maintain uniform output dimensions. A module, such as a residual block, serves as a fundamental building unit, comprising one or more layers designed to facilitate model growth and adaptation.

Based on this architectural characteristic, FedBridge trains *expandable* fusion models on edge servers instead of fixed-size architectures. Fig. 8 illustrates the expansion of a fusion model in FedBridge. At the beginning of the FL process, each fusion model is initialized with a shallow backbone network that comprises a fixed number of sub-networks, each containing only a single module. As training progresses, the fusion models expand until a predefined stopping criterion is met. During each FL training round, edge servers train their fusion models based on the ensemble knowledge from the client models through multiple training epochs. A critical challenge in this model expansion process is to determine when to expand a fusion model and when to stop. An effective method is to validate the model accuracy on labeled samples. If the accuracy improvement is less than ϵ over the last K epochs, the fusion model grows a module. If this model growth cannot improve model accuracy further by more than ϵ over the next K epochs, the network expansion stops for this training round.

However, model validation is infeasible in scenarios where edge servers lack access to proper validation data for model accuracy assessment. To overcome this limitation, FedBridge introduces an adaptive architectural expansion mechanism based on model stability monitoring. During the training of a fusion model, edge servers compute the L2-norm distance between fusion model parameters

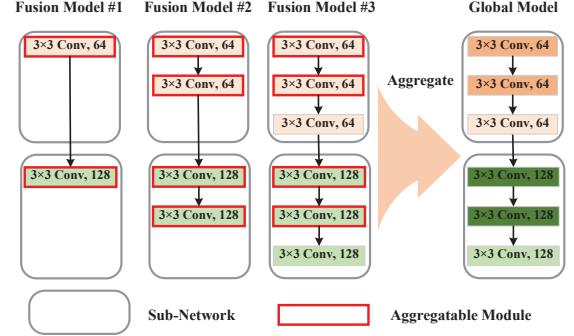


Figure 9: Fusion models aggregated into global model.

for consecutive epochs, i.e., $\|w_I^k - w_I^{k-1}\|_2$. If the distance remains below the threshold ϵ for K consecutive epochs, the fusion model automatically adds a module for each sub-network in the next FL training round. The distance is calculated as follows:

$$\frac{1}{K} \sum_{k=t-K+1}^t \|w_I^k - w_I^{k-1}\|_2 \leq \epsilon \quad (5)$$

When a fusion model expands, FedBridge monitors the L2-norm distance between its parameters (w_I^t) and the initial model parameters (w_I^0). If this distance remains below a predefined threshold ϵ , the fusion model stops further growth for the rest of the training round. To initialize a new module, FedBridge duplicates the parameters of the current modules on the same sub-network. This ensures continuity in parameter initialization and preserves previously learned knowledge.

4.3 Cloud Model Aggregation

In Step 4, the cloud server aggregates the fusion models received from the edge servers to construct the global model, which is then distributed back to the edge servers in Step 5. The model expansion mechanism employed by the edge servers in Step 2 diversifies the architectures and sizes of their fusion models. A straightforward approach for aggregating the knowledge in these heterogeneous fusion models is to utilize ensemble logits generated by them on unlabeled data. This aggregation process mirrors the procedure used for training the fusion models. However, compared to parameter averaging, ensemble knowledge distillation faces the limitation of insufficient knowledge fusion, as shown and discussed in Section 2.2. Although fusion models tend to become heterogeneous as the learning process proceeds, the modules within the corresponding sub-networks in these models share identical architectures. Leveraging this characteristic, FedBridge utilizes a block-based aggregation strategy for the cloud server to effectively integrate fusion models into a unified global model. Fig. 9 illustrates this aggregation process. The cloud server first identifies the common modules across the sub-networks of all fusion models and then aggregates their parameters by averaging them into the corresponding modules of the global model. Next, the cloud server transmits the model parts corresponding to individual edge servers for updating their fusion models. Algorithm 1 of Appendix A presents the pseudocode of the training process.

Table 2: Full comparison of top-1 accuracy.

Method	CIFAR-10					CINIC-10				
	IID		Lower Heterogeneity		Higher Heterogeneity	IID		Lower Heterogeneity		Higher Heterogeneity
	$S = 10$	$S = 5$	$\beta = 1$	$S = 2$	$\beta = 0.1$	$S = 10$	$S = 5$	$\beta = 1$	$S = 2$	$\beta = 0.1$
FedAvg	0.647	0.609	0.612	0.502	0.488	0.526	0.474	0.466	0.384	0.379
FedMD	0.559	0.502	0.514	0.393	0.362	0.482	0.416	0.392	0.323	0.298
FedDF	0.628	0.579	0.601	0.419	0.412	0.503	0.431	0.422	0.358	0.346
Fed-ET	0.637	0.594	0.618	0.464	0.428	0.521	0.472	0.457	0.364	0.355
AdaptiveFL	0.651	0.597	0.615	0.482	0.477	0.534	0.483	0.476	0.402	0.406
FedEcover	0.649	0.583	0.607	0.472	0.452	0.531	0.469	0.462	0.371	0.388
FedBridge	0.699	0.678	0.683	0.632	0.617	0.578	0.553	0.546	0.528	0.532

5 Evaluation

5.1 Experimental Setup

Implementation. FedBridge is evaluated experimentally in an edge-assisted FL system consisting of five physical machines functioning as edge servers within a private data center, with 50 clients randomly distributed across these edge servers. The cloud server operates on an Amazon EC2 c5.2xlarge instance with 8 vCPUs and 128GB of memory. Each client is equipped with an Intel i7-12700 CPU and 4GB, 8GB, or 16GB of memory.

Datasets, Models and Baselines. The models are trained on CIFAR-10 [23, 26] and CINIC-10 [6, 15], two datasets widely used in studies of on-device model training [33, 47, 53]. For non-IID settings in FedBridge. The first setting involves quantity-based label imbalance, where each client possesses data containing a fixed number of classes, denoted as S . The second setting is distribution-based label imbalance, where each client holds a proportion of labels following a Dirichlet distribution with concentration parameter β . A CNN [37] is assigned to each low-capacity client, a ResNet-8 [16] to each medium-capacity client, and a DeiT-Tiny [42] to each high-capacity client. This ensures that the model complexity aligns with the computational capacities of the clients. The CNN model comprises 4 layers with 0.2M parameters (0.8MB). ResNet-8 increases the depth to 8 layers with 1.1M parameters (4.1MB). DeiT-Tiny, also with 8 layers, contains 2.1M parameters with a size of 8.4MB. FedBridge is compared against FedAvg [37], FedMD [28], FedDF [34], Fed-ET [3], AdaptiveFL [21] and FedEcover [43]. And more evaluations are presented in Appendix B and Appendix C.

5.2 Overall Comparison

Top-1 Accuracy. Table 2 compares FedBridge and the baselines in top-1 accuracy across three different data distributions on CIFAR-10 and CINIC-10, where $S = 2, 5$, and 10 denotes that each client has 2, 5, and 10 classes of samples, and $\beta = 0.1$ and 1 denotes that each client has a proportion of the labels following a Dirichlet distribution with a concentration parameter of $\beta = 0.1$ and $\beta = 1$. There are three key observations.

- Compared to FedMD, FedDF, and Fed-ET, FedAvg achieves superior top-1 accuracy. For example, on CIFAR-10, FedAvg outperforms Fed-ET by 1.0%, 1.5%, 1.1%, 3.8%, and 6.0% under conditions " $S = 10$ ", " $S = 5$ ", " $\beta = 1$ ", " $S = 2$ ", and " $\beta = 0.1$ ", respectively. These results indicate that KD-based approaches indeed face challenges in effectively aggregating knowledge from heterogeneous client models into the global model, as discussed in Section 2.2.

- Fed-ET outperforms FedDF. For example, on CIFAR-10, Fed-ET achieves an accuracy improvement of 0.9%, 1.5%, 1.7%, 4.5%, and 1.6% under the conditions " $S = 10$ ", " $S = 5$ ", " $\beta = 1$ ", " $S = 2$ ", and " $\beta = 0.1$ ", respectively. This performance gain is primarily due to the ResNet-110 model utilized by Fed-ET on the cloud server, which is 3.6x larger than the ResNet-32 model employed by FedDF. Compared with ResNet-32, ResNet-110 allows Fed-ET to integrate the knowledge from heterogeneous client models more effectively. However, when we replace ResNet-32 with ResNet-110 for FedDF, there is only a 0.68% improvement in model accuracy. This validates the design of FedBridge to expand fusion models on edge servers rather than deploying large fixed-size fusion models (§4.2) for block-based aggregation (§4.3).
- FedBridge demonstrates superior performance in mitigating data heterogeneity compared to the baselines. For instance, when data heterogeneity increases with a decrease in S from 10 to 5, Fed-ET suffers a significant model accuracy drop of 4.3% and 4.9% on CIFAR-10 and CINIC-10, respectively. With its contrastive distillation mechanism (§4.2), FedBridge manages to reduce this model accuracy drop by 51.16% on CIFAR-10 and 48.80% on CINIC-10.
- Among all five systems, FedBridge achieves the highest model accuracy, outperforming the baselines by 4.8% - 14.0% on CIFAR-10 and 4.4% - 9.6% on CINIC-10, respectively. This highlights its effectiveness in fusing knowledge across heterogeneous client models. Compared to the Fed-ET, the state-of-the-art solution, FedBridge achieves an accuracy advantage of 6.2% on CIFAR-10 and 5.6% on CINIC-10.

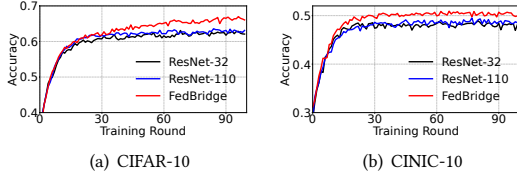
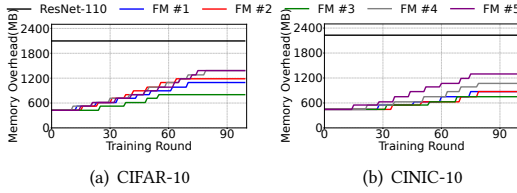
5.3 Ablation Study

Contrastive Loss. FedBridge incorporates a logit-contrastive loss during the training of fusion models to mitigate data heterogeneity (§4.2). To assess its usefulness, we conduct an experiment to compare the performance of FedBridge with and without the logit-contrastive loss for training fusion models. Table 3 compares the accuracy achieved, where there are 50 clients and 5 edge servers in the system. With logit-contrastive loss, FedBridge achieves an accuracy improvement of 2.4% on CIFAR-10 and 3.0% on CINIC-10. This indicates that the logit-contrastive loss indeed allows FedBridge to prevent biased training for fusion models introduced by data heterogeneity. It also accelerates model convergence.

Fusion Model Expansion. FedBridge deploys and trains expandable fusion models on edge servers to properly integrate the knowledge from heterogeneous client models (§4.2). To evaluate the usefulness of model expansion, we conduct an experiment where FedBridge trains expandable and fixed-size fusion models on edge

Table 3: FedBridge's top-1 accuracy with and without a logit contrastive losses on CIFAR-10 and CINIC-10.

Dataset	Fusion Model Training	
	without	with Contrastive Loss
CIFAR-10	0.654	0.678 (+3.67%)
CINIC-10	0.523	0.553 (+5.74%)

**Figure 10: Model convergence and accuracy with expandable and fixed-size fusion models in FedBridge, where there are 20 clients and 5 edge servers in the system, "ResNet-32" and "ResNet-110" denotes the fixed-sized fusion models, "FedBridge" denotes that the fusion models are expandable in the system.****Figure 11: Sizes of five fusion models (FM) as they expand over 100 training rounds. Black lines represent fixed-size ResNet-110.**

servers. Table 4 shows the results. We can see that model expansion improves model accuracy by 4.0% and 2.6% on CIFAR-10, when fixed-size ResNet-32 and ResNet-110 are deployed on edge servers. Similarly, on CINIC-10, expandable fusion models achieve an accuracy gain of 2.1% and 1.6% against fixed-size ResNet-32 and ResNet-110, respectively. Fig. 10 depicts the convergence of the global model when FedBridge deploys expandable and fixed-size fusion models on edge servers for knowledge fusion. The comparison tells us that with expandable fusion models, FedBridge takes 35.3% less time to converge ResNet-32 on CIFAR-10 and 77.5% less time to converge ResNet-32 on CINIC-10. In the meantime, FedBridge also reduces the memory overhead on edge servers with expandable fusion models on demand rather than fixed-size ones. Fig. 11 demonstrates memory overheads incurred when FedBridge trains the five expandable fusion models as they grow over 100 training rounds in this experiment. We can see that they each go through an average of 7 expansions to accommodate the increasing knowledge extracted from client data throughout the entire FL process before they stabilize. We can also observe their memory efficiency as compared to the use of fixed-size ResNet-110. In total, training these five expandable models incurs only 48.12% of the memory overhead incurred by training their fixed-size ResNet-110 counterparts. This is a profound advantage of FedBridge in practice as it allows fusion models to grow further over time to accommodate more knowledge.

Table 4: Top-1 accuracy with expandable and fixed-size fusion models in FedBridge on CIFAR-10 and CINIC-10, where "ResNet-32" and "ResNet-110" denote the fixed-size fusion models, and "Expandable" denotes that the fusion models are expandable.

Dataset	Fusion Model		
	ResNet-32	ResNet-110	Expandable
CIFAR-10	0.631 (-6.34%)	0.645 (-4.03%)	0.671
CINIC-10	0.486 (-4.32%)	0.491 (-3.26%)	0.507

Table 5: Top-1 accuracy of FedBridge with KD-based and aggregation-based knowledge fusion on the cloud server with a system comprising 20 clients and 5 edge servers.

Dataset	Knowledge Fusion in Cloud	
	KD-based	Aggregation-based (FedBridge)
CIFAR-10	0.583	0.671 (+15.10%)
CINIC-10	0.476	0.507 (+6.51%)

Knowledge Distillation vs. Model Aggregation. As discussed in Section 2, compared with knowledge distillation, model aggregation is more effective in knowledge aggregation across multiple models. To integrate the knowledge from heterogeneous fusion models produced by model expansion, FedBridge averages their shared modules into the global model on the cloud server. To assess the usefulness of this design, we conduct an experiment to observe its performance when it employs knowledge distillation on the cloud server to integrate the knowledge from fusion models. Table 5 shows the results. When FedBridge integrates knowledge from fusion models through knowledge distillation rather than model aggregation, it suffers a model accuracy loss of 8.8% and 3.1% on CIFAR-10 and CINIC-10, respectively. Fig. 12 in Appendix B illustrates the corresponding model convergence processes. It can be seen that model convergence is 5.64x slower. The results validate our novel design for model aggregation on the cloud server.

6 Conclusion

This paper presented FedBridge, a novel edge-assisted FL system that allows model-heterogeneous clients to share their knowledge communication-efficiently in a federated manner. On edge servers, it deploys and trains expandable fusion models instead of fixed-size models to integrate the knowledge extracted from client models effectively and memory-efficiently. In non-IID settings, FedBridge accommodates the heterogeneity in client data with a new contrastive distillation technique when training the fusion models. On the cloud server, it merges heterogeneous fusion models through aggregation instead of knowledge distillation. Compared to state-of-the-art edge-assisted FL systems, FedBridge accelerates model convergence by up to 6.3x times and improves model accuracy by up to 17.7%, with an average reduction of 92.82% in communication overhead and 54.95% in memory overhead.

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A Algorithm

Algorithm 1: FedBridge Training

```

1 Function FusionModelTraining()
2   For client model  $w_L$ 
3      $z_L \leftarrow f(w_L; D^c)$ 
4    $z_G \leftarrow f(w_G; D^c)$ 
5   For epoch  $e = 0, 1, 2, \dots, E$ 
6      $z_I^e \leftarrow f(w_I^e; D^c)$ 
7      $L_{KD} \leftarrow KL\left(\sigma\left(\frac{1}{|S|} \sum_{u \in S} z_L^u\right), \sigma\left(z_I^e\right)\right)$ 
8      $L_{CD} \leftarrow$ 
9        $-\sum_{u \in S} \frac{1}{|S|} \log\left(\frac{\exp(\text{sim}(z_I^e, z_G)/\tau)}{\exp(\text{sim}(z_I^e, z_G)/\tau) + \exp(\text{sim}(z_I^e, z_L^u)/\tau)}\right)$ 
10     $L \leftarrow L_{KD} + \mu L_{CD}$ 
11     $w_I^e \leftarrow w_I^e - \eta \nabla L$ 
12    If  $e == E$  then
13       $D = \frac{1}{K} \sum_{k=E-K+1}^E \|w_I^k - w_I^{k-1}\|_2$ 
14      If  $D \leq \epsilon$  then
15        expand a module for each sub-network in
16        the next round
17      If  $e == K$  then
18        If  $w_I$  expanded in the last round then
19           $D = \|w_I^0 - w_I^K\|_2$ 
20          If  $D \leq \epsilon$  then
21            stop expanding the fusion model in the
22            rest of the rounds
23    Send  $w_I$  to the cloud server.

```

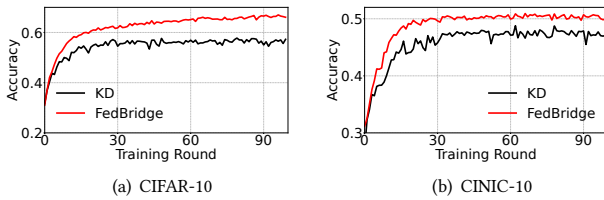


Figure 12: Comparison of global model convergence between averaged aggregation and knowledge distillation on the cloud server in FedBridge, with a system comprising 20 clients and 5 edge servers. "KD" denotes the knowledge distillation-based aggregation method, while "FedBridge" denotes the averaged aggregation approach.

B Extra Experiments

Training Rounds and Communication Overheads. Table 6 compares the number of training rounds required and communication overheads incurred for FedBridge and the baselines to achieve the target accuracy on CIFAR-10 and CINIC-10. It shows FedBridge needs the fewest training rounds to convergence a model to the target accuracy and incur the least communication overhead in the process. For example, FedBridge needs only 22 and 26 training rounds to converge an accuracy of 62.8% and 50.3% on CIFAR-10 and

CINIC-10 while Fed-ET needs 139 and 152 training rounds to do the same. Its communication overhead is also much lower, i.e., a total of 359 MB and 425 MB vs. 9,285 MB and 10,153 MB on CIFAR-10 and CINIC-10, respectively. The high training efficiency of FedBridge is a great advantage because it allows the system to accommodate a large number of clients without a traffic overhead concern. High data diversity offered by these clients, in turn, improves model accuracy, as demonstrated in Fig. 13.

Table 6: Full comparison of training rounds and communication overhead (MB) when the global model reaches a target accuracy, i.e., 62.8% on CIFAR-10 and 50.3% on CINIC-10 in IID setting. Note: FedMD failed to convert the global model to the target accuracy.

Method	CIFAR-10		CINIC-10	
	Rounds	Overhead	Rounds	Overhead
FedAvg	89	3,560	82	3,280
FedMD	N/A	N/A	N/A	N/A
FedDF	171	11,422	183	12,224
Fed-ET	139	9,285	152	10,153
AdaptiveFL	85	1,710	78	1,605
FedEcover	92	2,852	85	2,418
FedBridge	22	359	26	425

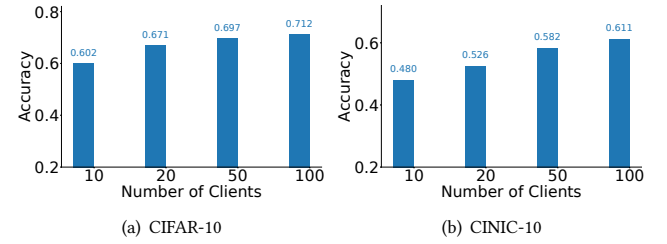


Figure 13: Top-1 accuracy with varying numbers of clients in FedBridge.

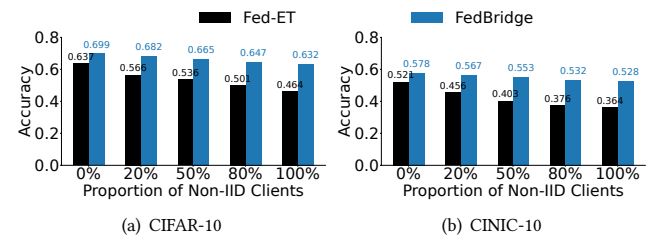


Figure 14: Comparison in top-1 accuracy between FedBridge and Fed-ET across varying proportions of non-IID clients, with a system comprising 20 clients and 5 edge servers.

Top-1 accuracy. Fig. 13 show the Top-1 accuracy with varying numbers of clients in FedBridge. It shows High data diversity offered by participating clients improves model accuracy.

Global model convergence and accuracy. Fig. 12 illustrates the model convergence process. The convergence is observed to be $5.64\times$ slower, which further validates the effectiveness of our FedBridge design for model aggregation on the cloud server.

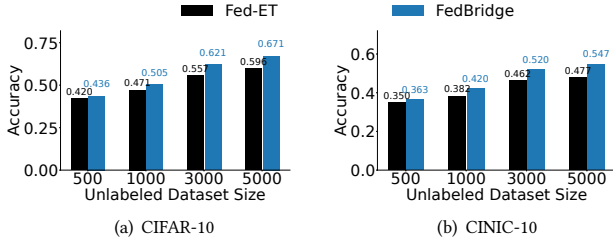


Figure 15: Top-1 accuracy comparison between FedBridge and Fed-ET across varying unlabeled dataset sizes, with a system comprising 20 clients and 5 edge servers.

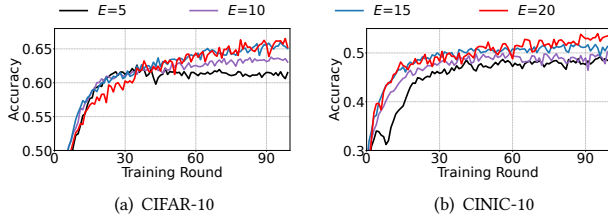


Figure 16: Model convergence and accuracy with different numbers of epochs for training fusion models, with 20 clients across 5 edge servers in the system.

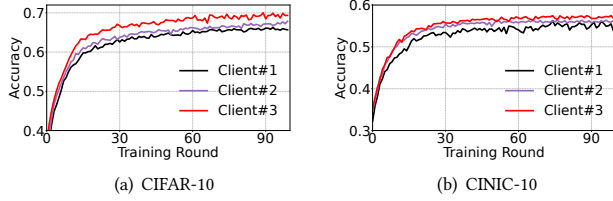


Figure 17: Top-1 accuracy of FedBridge under different client configurations, where there are 50 clients and 5 edge servers in the system.

C In-Depth Evaluation

This section assesses the performance of FedBridge in various edge-assisted FL scenarios.

Impact of μ . As introduced in Section 4.2, FedBridge employs a hyperparameter μ to adjust the contribution of contrastive distillation loss to fusion model training. To evaluate its impact, we evaluate the performance of FedBridge with $\mu = 0.01, 0.1, 1.0$. Table 7 summarizes the results. It shows that FedBridge achieves optimal performance with $\mu = 0.1$ on CINIC-10 and $\mu = 0.1$ on CINIC-10.

Table 7: FedBridge’s top-1 accuracy with different μ values.

Dataset	Fusion Model Training		
	$\mu = 0.01$	$\mu = 0.1$	$\mu = 1$
CIFAR-10	0.657	0.678	0.661
CINIC-10	0.538	0.553	0.544

Impact of Proportion of Non-IID Clients. As discussed in Section 2.2, an increasing proportion of non-IID clients poses challenges for model convergence in KD-based FL systems such as

FedDF and Fed-ET. To evaluate the effectiveness of FedBridge in mitigating data heterogeneity, we vary the proportion of non-IID clients and compare its performance against Fed-ET. The results, as shown in Fig. 14, demonstrate that Fed-ET suffers a significant accuracy drop of 17.3% on CIFAR-10 and 15.7% on CINIC-10, while FedBridge reduces this accuracy drop by 61.27% and 68.15%, respectively.

Impact of Unlabeled Data. FedBridge trains fusion models based on the knowledge from client models through knowledge distillation, which requires unlabeled data (§4.2). The amount of unlabeled data has a potential impact on model convergence and accuracy. To investigate its impact, we vary the number of unlabeled data samples from 500 to 5,000 and evaluate FedBridge against Fed-ET, which also relies on a public dataset. Fig. 15 shows the results. FedBridge achieves an increasingly significant performance improvement when the amount of unlabeled data increases. For example, FedBridge achieves an accuracy improvement of 23.5% and 15.4%, when the amount of unlabeled data increases from 500 to 5,000 for CIFAR-10 and CINIC-10, respectively, while Fed-ET only obtains an accuracy improvement of 15.6% and 10.7%. This indicates that compared with Fed-ET, FedBridge generates more accurate logits from unlabeled data as supervisory information for both training fusion models and client models.

Impact of Number of Epochs for Training Fusion Models. FedBridge trains fusion models on edge servers, where the number of epochs E for training these fusion models may impact its training performance. In this experiment, we evaluate the impact of E on model convergence and accuracy by varying its value from 5 to 20. The results, presented in Fig. 16, demonstrate that an increase in E improves both model accuracy and convergence for FedBridge. It achieves an accuracy improvement of 4.9% and 4.7% when E increases from 5 to 20 in the experiments on CIFAR-10 and CINIC-10, respectively. It demonstrates that a larger E allows edge servers to more effectively integrate knowledge from heterogeneous client models, which improves model accuracy and accelerates model convergence.

Impact of Client Capacities. As discussed in Section 2.1, model heterogeneity can be caused by the heterogeneity in client capacities. For example, a high-capacity client may afford to run ViT while a low-capacity client can only afford a CNN. This experiment evaluates the impact of client capacities on the performance of FedBridge. We consider three client configurations: (C1) 50% low-capacity clients, 20% medium-capacity clients, and 30% high-capacity clients; (C2) 20% low-capacity clients, 50% medium-capacity clients, and 30% high-capacity clients; and (C3) 20% low-capacity clients, 30% medium-capacity clients, and 50% high-capacity clients. The results, presented in Fig. 17, show that FedBridge achieves an accuracy of 69.9% and 57.8% on CIFAR-10 and CINIC-10, respectively, in C#3, 4.1% and 2.2% higher than C#1. This shows that clients with larger models can extract more knowledge from their training data, unsurprisingly. More importantly, it indicates that FedBridge can expand its fusion models properly to accommodate the extra knowledge extracted, which translates to its significant performance gains.